Low-Rank Tensor Approximation for Image Inpainting: A Comparative Study of Decomposition Methods, Optimization Algorithms, and Reconstruction Norms

Mathematical Analysis and Numerical Experiments on Norm-dependent Reconstruction Behavior

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Project Information	
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Revision History

The following table summarizes the major revisions and updates applied to this document.

Version	Date	Author	Description
0.1	2026-05-21	Beomjun Chung	Initial structure
0.2	2026-06-03	Beomjun Chung	Fix abstract, appendix; add figure references

Table 1: Revision history of the document

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Abstract

This report presents a comparative study of low-rank tensor approximation methods applied to colour image reconstruction, covering both image compression and robust recovery from corrupted observations. Colour images are modelled as third-order tensors of shape $H \times W \times 3$, and low-rank structure is exploited by two decomposition families: Tucker decomposition, parameterised by a multi-linear rank (r_1, r_2, r_3) and a compact core tensor, and CP (CANDECOMP/PARAFAC) decomposition, which expresses the tensor as a weighted sum of rank-1 outer products.

Three reconstruction objectives are investigated: the squared Frobenius norm, which admits analytically tractable Alternating Least Squares (ALS) updates and is optimal under Gaussian noise; the ℓ_1 norm, which is robust to sparse outliers and is minimised via Iteratively Reweighted Least Squares (IRLS); and the Huber loss, which smoothly interpolates between the two and is optimised with Adam gradient descent. Tucker decomposition is implemented from scratch using Higher-Order SVD (HOSVD) initialisation followed by Higher-Order Orthogonal Iteration (HOOI) refinement, with correctness verified against TensorLy.

Four controlled experiments validate the theoretical framework. First, the empirical HOSVD reconstruction error satisfies the theoretical multilinear singular-value error bound for all tested ranks, with HOOI consistently improving upon the HOSVD baseline. Second, under sparse pixel corruption, ℓ_1-Tucker (IRLS) outperforms Frobenius-Tucker (HOOI) by a margin that grows with corruption density; Frobenius-Tucker remains optimal under Gaussian noise. Third, HOOI converges in approximately 10–20 iterations whereas Adam requires several hundred, confirming the efficiency advantage of closed-form ALS updates. Fourth, HOSVD initialisation achieves lower and zero-variance reconstruction error compared with random initialisation, reducing the impact of non-convexity.

Keywords: Tucker Decomposition, CP Decomposition, HOSVD, HOOI, Iteratively Reweighted Least Squares, Image Compression, Robust Tensor Approximation, Frobenius Norm, ℓ_1 Norm, Huber Loss

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1. Introduction

Digital images are conventionally stored and processed as three-dimensional arrays of shape $H \times W \times C$, where H and W denote the spatial dimensions and C the number of color channels. Although matrix-based compression methods such as truncated Singular Value Decomposition (SVD) are well-established, they require either flattening the channel axis or processing each channel independently, thereby discarding inter-channel correlations that carry perceptually significant structure.

Tensor decomposition methods naturally preserve this multi-way structure. A *tensor* $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ generalizes matrices to N modes, and *low-rank tensor approximation* seeks a compact representation $\hat{\mathcal{X}}$ such that

$$\hat{\mathcal{X}} = \arg\min_{\hat{\mathcal{X}} \in \mathcal{M}_r} \mathcal{L}\Big(\mathcal{X}, \hat{\mathcal{X}}\Big), \tag{1.1}$$

where $\mathcal{M}_r$ denotes a suitable low-rank manifold and $\mathcal{L}$ is a reconstruction loss. The choice of $\mathcal{L}$ is consequential: the squared Frobenius norm $\|\cdot\|_F^2$ yields analytically tractable subproblems and is the standard baseline, yet it is sensitive to outliers and heavy-tailed noise. Alternative losses—the ℓ_1 norm and the Huber loss—offer robustness at the cost of algorithmic complexity, motivating a comparative study.

Research Questions

This project investigates the following questions:

1. **Decomposition structure.** How do Tucker decomposition and CP decomposition differ in their approximation quality, compression ratio, and computational cost for image tensors?

2. **Norm dependence.** How does the choice of reconstruction norm ($\|\cdot\|_F^2$, $\|\cdot\|_1$, Huber loss) affect the recovered factor matrices and the visual quality of the reconstructed image, particularly under structured noise?

3. **Optimization behavior.** How do Alternating Least Squares (ALS) / Higher-Order Orthogonal Iteration (HOOI) compare with gradient-based methods (Adam, L-BFGS) in terms of convergence speed, stability, and sensitivity to initialization?

4. **Rank–quality tradeoff.** Can the theoretical error bound derived from the multilinear singular value spectrum predict the empirically observed reconstruction error as a function of Tucker rank?

Contributions

The contributions of this work are as follows.

- We implement Tucker decomposition from scratch (HOSVD initialization and HOOI refinement) and verify correctness against TensorLy [4].

- We derive and implement an Iteratively Reweighted Least Squares (IRLS) algorithm for ℓ_1-Tucker and compare it with the Frobenius baseline and Huber-loss gradient descent on noisy image tensors.

- We conduct controlled numerical experiments on rank truncation, norm sensitivity, convergence, and initialization, connecting each empirical result to its theoretical prediction.

- We demonstrate a lightweight image compression pipeline that reports PSNR and compression ratio as functions of Tucker rank, providing an interpretable application of the mathematical framework.

Paper Organization

The remainder of this report is organized as follows. Section 2 establishes the mathematical background, including tensor notation, Tucker and CP decomposition, and the norms used throughout. Section 3 derives the optimization algorithms: ALS/HOOI for the Frobenius objective and IRLS plus gradient-based methods for robust objectives. Section 4 analyzes the statistical and geometric implications of each norm choice. Section 5 presents numerical experiments that validate the theoretical predictions. Section 6 applies the framework to image compression. Section 7 interprets the results and discusses limitations, and Section 8 concludes.

2. Mathematical Background

This section establishes the notation and mathematical foundations used throughout the report. We introduce tensor algebra, the two decomposition families under study (Tucker and CP), the multilinear singular value decomposition (HOSVD), and the reconstruction losses that define each optimization problem.

2.1. Tensor Notation

A *tensor* $X \in^{I_1 \times I_2 \times \cdots \times I_N}$ is an N-way array. Its entry indexed by $(i_1, i_2, \ldots, i_N)$ is written $x_{i_1 i_2 \cdots i_N}$. We adopt the following notational convention throughout: calligraphic letters (X) for tensors, bold uppercase (U) for matrices, bold lowercase (u) for vectors, and italic lowercase (λ, r) for scalars.

Frobenius norm. The Frobenius norm of a tensor generalizes the matrix Frobenius norm entry-wise:

$$X \;=\; \sqrt{\sum_{i_1=1}^{I_1} \cdots \sum_{i_N=1}^{I_N} x_{i_1 \cdots i_N}^2} \;=\; \sqrt{\langle X,\, X \rangle}. \tag{2.1}$$

It satisfies *unitary invariance*: for any orthogonal matrices Q_n,

$$X \times_1 Q_1 \times_2 Q_2 \cdots \times_N Q_N \;=\; X. \tag{2.2}$$

This property is central to the correctness of HOSVD and HOOI (Section 2.4).

Mode-n unfolding (matricization). The *mode-n unfolding* $Xn \in^{I_n \times \prod_{k \neq n} I_k}$ rearranges the entries of X such that the mode-n index maps to rows and all remaining indices (in lexicographic order) map to columns [3]. Concretely,

$$\big[Xn\big]_{i_n,\, j} \;=\; x_{i_1 i_2 \cdots i_N}, \qquad j = 1 + \sum_{\substack{k=1 \\ k \neq n}}^{N} (i_k - 1) \prod_{\substack{m=1 \\ m \neq n}}^{k-1} I_m. \tag{2.3}$$

Mode-n product. The *mode-n product* of $X \in^{I_1 \times \cdots \times I_N}$ with $U \in^{J \times I_n}$ is the tensor $Y = X \times_n U \in^{I_1 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_N}$ defined entry-wise as

$$y_{i_1 \cdots i_{n-1}\, j\, i_{n+1} \cdots i_N} \;=\; \sum_{i_n=1}^{I_n} x_{i_1 \cdots i_N}\, u_{j i_n}. \tag{2.4}$$

In matricized form this becomes

$$\big(X \times_n U\big)_{(n)} \;=\; U\, Xn, \tag{2.5}$$

which reduces the mode-n product to ordinary matrix multiplication and underpins the ALS update rules derived in Section 3.1.

Commutativity of mode products. For $m \neq n$,

$$X \times_m A \times_n B \;=\; X \times_n B \times_m A. \tag{2.6}$$

This commutativity is used repeatedly in the derivation of ALS subproblems.

2.2. Tucker Decomposition

[Tucker decomposition] A rank-$(r_1, \dots, r_N)$ Tucker approximation of $X \in^{I_1\times\cdots\times I_N}$ is

$$\hat{X} \;=\; G \times_1 U^{(1)} \times_2 U^{(2)} \cdots \times_N U^{(N)}, \tag{2.7}$$

where $G \in^{r_1\times\cdots\times r_N}$ is the *core tensor* and each factor matrix $U^{(n)} \in^{I_n\times r_n}$ has orthonormal columns, i.e., ${U^{(n)}}^\top U^{(n)} = I_{r_n}$.

Mode-n unfolding of $\hat{X}$ yields

$$\hat{X}_{(n)} \;=\; U^{(n)}\, G_{(n)} \Big(U^{(N)} \cdots U^{(n+1)} U^{(n-1)} \cdots U^{(1)}\Big)^\top, \tag{2.8}$$

where denotes the Kronecker product. Equation (2.8) exposes the least-squares structure exploited by ALS.

Parameter count and compression ratio. The Tucker format requires

$$P_{\text{Tucker}} \;=\; \prod_{n=1}^{N} r_n \;+\; \sum_{n=1}^{N} I_n r_n \tag{2.9}$$

parameters. The compression ratio relative to the dense tensor is

$$\rho \;=\; \frac{\prod\limits_{n=1}^{N} I_n}{P_{\text{Tucker}}}. \tag{2.10}$$

For a three-channel image of size $H \times W \times 3$ with Tucker rank $(r, r, 3)$, this gives $\rho = HW \cdot 3\,/\,(3r^2 + Hr + Wr + 3r)$, which grows roughly as HW/r^2 for large images.

Relationship to truncated SVD. For $N = 2$, Tucker reduces to the truncated SVD: $\hat{X} = U\Sigma_r V^\top$, with $G = \Sigma_r$ and $U^{(1)} = U$, $U^{(2)} = V$. Tucker thus provides the natural multilinear generalization of low-rank matrix approximation.

2.3. CP Decomposition

[CP decomposition] A rank-R CANDECOMP/PARAFAC (CP) approximation of X is

$$\hat{X} \;=\; \sum_{r=1}^{R} \lambda_r\, a_r^{(1)} \circ a_r^{(2)} \circ \cdots \circ a_r^{(N)}, \tag{2.11}$$

where $\circ$ denotes the outer product, $\|a_r^{(n)}\|_2 = 1$, and $\lambda_r \geq 0$.

CP can be viewed as Tucker with a *superdiagonal* core of size $R \times \cdots \times R$, i.e., the case $r_1 = \cdots = r_N = R$ with $g_{i_1\cdots i_N} = \lambda_{i_1}$ only when $i_1 = i_2 = \cdots = i_N$ and zero otherwise. The CP format stores $R\big(\sum_{n=1}^{N} I_n + 1\big)$ parameters, fewer than Tucker for the same R, but determining the exact CP rank is NP-hard in general [3].

[CP degeneracy] Unlike Tucker, the best rank-R CP approximation of a tensor of order $N \geq 3$ may not exist: the infimum of the objective is not always attained [3]. Tucker decomposition avoids this pathology because orthonormal factor matrices form a compact set.

2.4. Higher-Order SVD (HOSVD)

[HOSVD [1]] The *Higher-Order SVD* of X computes each factor matrix independently:

$$U^{(n)} \;:=\; \text{leading } r_n \text{ left singular vectors of } Xn, \quad n = 1, \ldots, N, \tag{2.12}$$

and sets the core tensor to

$$G \;=\; X \times_1 U^{(1)^\top} \times_2 U^{(2)^\top} \cdots \times_N U^{(N)^\top}. \tag{2.13}$$

HOSVD is computed in closed form and provides a deterministic initialization. However, it is *not* the globally optimal Tucker approximation because the mode-n factor matrices are optimized independently rather than jointly.

[HOSVD error bound [1]] Let $\hat{X}_{\text{HOSVD}}$ denote the rank-$(r_1, \ldots, r_N)$ HOSVD approximation of X. Then

$$X - \hat{X}_{\text{HOSVD}}{}^2 \;\leq\; \sum_{n=1}^{N} \sum_{i=r_n+1}^{I_n} \sigma_i^{(n)2}, \tag{2.14}$$

where $\sigma_i^{(n)}$ denotes the i-th singular value of Xn.

Equation (2.14) is used in Section 5 to benchmark the rank sweep experiment: the left-hand side is measured empirically and compared with the right-hand side computed from the singular value spectra.

2.5. Reconstruction Losses

The general approximation problem is

$$\min_{G,\,\{U^{(n)}\}} \; \mathcal{L}\Big(X,\, \hat{X}\Big), \tag{2.15}$$

where the choice of $\mathcal{L}$ determines both the statistical interpretation of the solution and the algorithmic complexity of the optimization. We study three loss functions.

Squared Frobenius loss.

$$\mathcal{L}_F(X, \hat{X}) \;=\; X - \hat{X}^2 \;=\; \sum_{i_1,\ldots,i_N} \big(x_{i_1\cdots i_N} - \hat{x}_{i_1\cdots i_N}\big)^2. \tag{2.16}$$

This loss is smooth and convex in each factor separately, enabling closed-form ALS updates. When the noise is i.i.d. Gaussian, the Frobenius minimizer is the maximum likelihood estimator. Its sensitivity to large residuals, however, makes it non-robust to outliers.

ℓ_1 loss.

$$\mathcal{L}_1(X, \hat{X}) \;=\; X - \hat{X} \;=\; \sum_{i_1,\ldots,i_N} |x_{i_1\cdots i_N} - \hat{x}_{i_1\cdots i_N}|. \tag{2.17}$$

Minimizing $\mathcal{L}_1$ corresponds to maximum likelihood estimation under a Laplace noise model and yields a positive breakdown point, making it robust to sparse outliers. The loss is convex but *non-smooth* at zero; its subdifferential is $\partial|\cdot| = (\cdot)$, which precludes direct use of standard least-squares solvers.

Huber loss. The Huber loss [2] interpolates smoothly between ℓ_2 (near zero) and ℓ_1 (in the tails):

$$\rho_\delta(r) \;=\; \begin{cases} \dfrac{1}{2} r^2 & \text{if } |r| \leq \delta, \\[2ex] \delta\Big(|r| - \dfrac{\delta}{2}\Big) & \text{if } |r| > \delta, \end{cases} \tag{2.18}$$

with threshold $\delta > 0$. The aggregate loss is $\mathcal{L}_H = \sum_{i_1,\ldots,i_N} \rho_\delta(x_{i_1\cdots i_N} - \hat{x}_{i_1\cdots i_N})$. Being everywhere differentiable, $\mathcal{L}_H$ admits gradient-based optimization without subgradient complications.

Summary. Table 2 summarizes the three losses with respect to statistical interpretation, smoothness, and optimization strategy.

Table 2: Comparison of reconstruction losses.

Loss	Noise model	Smoothness	Optimization
$\mathcal{L}_F$ (Frobenius)	Gaussian	C^∞	ALS (closed-form)
$\mathcal{L}_1$ (ℓ_1)	Laplace	non-smooth at 0	IRLS / subgradient
$\mathcal{L}_H$ (Huber)	Gaussian + outliers	C^1	Gradient descent

3. Optimization Methods

The Tucker approximation problem (2.15) is *jointly non-convex* in the factor matrices $\{U^{(n)}\}$ and the core tensor G. We study three algorithmic families that address this non-convexity in different ways: Alternating Least Squares (ALS) with its higher-order refinement HOOI for the Frobenius objective, Iteratively Reweighted Least Squares (IRLS) for the ℓ_1 objective, and gradient-based methods for the Huber objective.

3.1. Alternating Least Squares and HOOI

ALS principle. Although $\mathcal{L}_F$ is jointly non-convex in all factors simultaneously, it is *quadratic* — and hence convex — when all but one factor matrix are held fixed. ALS exploits this structure by cycling over the factors and solving each resulting least-squares problem exactly.

Mode-n subproblem. Fix all factor matrices except $U^{(n)}$ and define

$$Z^{(n)} \;:=\; G_{(n)}\Big(U^{(N)} \cdots U^{(n+1)} U^{(n-1)} \cdots U^{(1)}\Big)^{\top} \;\in\; {}^{r_n \times \prod_{k\neq n} I_k}. \tag{3.1}$$

The objective reduces to

$$\min_{U^{(n)}} \|Xn - U^{(n)} Z^{(n)}\|_F^2, \tag{3.2}$$

whose closed-form solution is

$$U^{(n)} \;\leftarrow\; XnZ^{(n)\top}\Big(Z^{(n)} Z^{(n)\top}\Big)^{-1}. \tag{3.3}$$

When the columns of $U^{(n)}$ are additionally constrained to be orthonormal — as in Tucker — the optimal solution is the matrix of leading r_n left singular vectors of $XnZ^{(n)\top}$, making the matrix inverse unnecessary.

HOOI algorithm. Higher-Order Orthogonal Iteration (HOOI) [?] applies ALS with orthonormal constraints. At each iteration, the contracted tensor

$$Y^{(n)} \;=\; X \times_1 U^{(1)^{\top}} \cdots \times_{n-1} U^{(n-1)^{\top}} \times_{n+1} U^{(n+1)^{\top}} \cdots \times_N U^{(N)^{\top}} \tag{3.4}$$

is formed (omitting mode n), and the update is

$$U^{(n)} \;\leftarrow\; \text{leading-}r_n \text{ left singular vectors of } Y^{(n)}_{(n)}. \tag{3.5}$$

After all factor matrices have converged, the core is recovered as

$$G \;=\; X \times_1 U^{(1)^{\top}} \times_2 U^{(2)^{\top}} \cdots \times_N U^{(N)^{\top}}. \tag{3.6}$$

Algorithm 1 summarizes the full procedure.

Convergence. Each HOOI iteration decreases the Frobenius reconstruction error monotonically, because each mode-n update is an exact minimizer of the subproblem (3.2) given the other factors. The sequence $\{e_t\}$ is therefore non-increasing and bounded below by zero, guaranteeing convergence of the *objective value*. However, convergence of the *iterates* $\{U_t^{(n)}\}$ to a stationary point is not guaranteed in general due to non-convexity [?].

Computational cost. The dominant cost per HOOI iteration is N truncated SVDs of matrices of size $I_n \times \prod_{k\neq n} r_k$, giving a per-iteration complexity of $O\Big(N \cdot I \cdot r^{N-1}\Big)$ for a balanced tensor with $I_n = I$ and $r_n = r$.

Algorithm 1 HOOI for Tucker decomposition (Frobenius norm)

Tensor $X \in^{I_1 \times \cdots \times I_N}$, target rank $(r_1, \ldots, r_N)$, tolerance $\varepsilon > 0$, max iterations T Core G, factor matrices $\{U^{(n)}\}$
Initialization via HOSVD

for $n \leftarrow 1$ N **do** $U^{(n)} \leftarrow$ leading-r_n left singular vectors of Xn Iterative refinement

for $t \leftarrow 1$ T **do**

for $n \leftarrow 1$ N **do** $Y \leftarrow X \times_{k \neq n} U^{(k)\top}$ $U^{(n)} \leftarrow$ leading-r_n left singular vectors of Yn $G \leftarrow X \times_1 U^{(1)\top} \cdots \times_N U^{(N)\top}$ $e_t \leftarrow X - G \times_1 U^{(1)} \cdots \times_N U^{(N)}$ $|e_t - e_{t-1}| < \varepsilon$**break return** G, $\{U^{(n)}\}$

3.2. Iteratively Reweighted Least Squares (IRLS) for ℓ_1-Tucker

Motivation. Direct minimization of $\mathcal{L}_1$ via subgradient methods converges at rate $O(1/\sqrt{t})$, which is slow in practice. IRLS sidesteps non-smoothness by solving a sequence of *weighted* Frobenius problems, each of which admits a standard ALS step.

IRLS reweighting. Let $r_{i_1 \cdots i_N} = x_{i_1 \cdots i_N} - \hat{x}_{i_1 \cdots i_N}$ be the current residual tensor. Define element-wise weights

$$w_{i_1 \cdots i_N}^{(t)} = \frac{1}{|r_{i_1 \cdots i_N}^{(t)}| + \varepsilon}, \quad \varepsilon > 0 \text{ (smoothing parameter)}, \tag{3.7}$$

and the *weighted* Frobenius problem

$$\min_{G, \{U^{(n)}\}} \sum_{i_1, \ldots, i_N} w_{i_1 \cdots i_N}^{(t)} \big(x_{i_1 \cdots i_N} - \hat{x}_{i_1 \cdots i_N}\big)^2. \tag{3.8}$$

The connection to ℓ_1 follows from the identity

$$|r| = w \cdot r^2 \quad \text{when } w = 1/|r|, \tag{3.9}$$

which shows that minimizing (3.8) with fixed weights is a majorization step for $\mathcal{L}_1$.

Weighted ALS update. Let $W^{(t)}$ be the weight tensor and define the element-wise product $\tilde{X} = W^{(t)} \odot X$. The weighted mode-$n$ subproblem is

$$\min_{U^{(n)}} \|\tilde{X}_{(n)} - U^{(n)} \tilde{Z}^{(n)}\|_F^2, \tag{3.10}$$

which is solved by (3.3) applied to the weighted unfolding $\tilde{X}_{(n)}$.

Algorithm 2 presents the full IRLS procedure.

Choice of ε. The parameter ε prevents division by zero and controls the approximation of $|\cdot|$ near the origin. Smaller ε gives a closer approximation to the true ℓ_1 loss but may cause numerical instability. A practical schedule starts with $\varepsilon = 10^{-1}$ and halves it every few outer iterations.

3.3. Gradient-Based Optimization for Huber-Tucker

Motivation. The Huber loss $\mathcal{L}_H$ is everywhere differentiable, enabling direct application of first- and second-order gradient methods. We use this objective to compare gradient-based optimization with ALS/IRLS in terms of convergence speed and final reconstruction quality.

Algorithm 2 IRLS for ℓ_1-Tucker decomposition

Tensor X, rank $(r_1, \ldots, r_N)$, outer iterations T, inner ALS steps K, smoothing $\varepsilon > 0$ Core G, factor matrices $\{U^{(n)}\}$
$G, \{U^{(n)}\} \leftarrow \mathrm{HOOI}(X, \mathbf{r})$ *warm start from Frobenius solution

for $t \leftarrow 1\ T$ **do** $\hat{X} \leftarrow G \times_1 U^{(1)} \cdots \times_N U^{(N)}$ $R \leftarrow X - \hat{X}$ *residual $W \leftarrow 1\,/\,(|R| + \varepsilon)$ *element-wise weights (3.7)

for $k \leftarrow 1\ K$ **do** One pass of weighted ALS

for $n \leftarrow 1\ N$ **do** $U^{(n)} \leftarrow$ weighted ALS update (3.10) $G \leftarrow X \times_1 U^{(1)\top} \cdots \times_N U^{(N)\top}$
return G, $\{U^{(n)}\}$

Gradient derivation. Let $R = X - \hat{X}$ be the residual and define the element-wise Huber pseudo-residual

$$\psi_\delta(r) \;=\; \frac{\partial \rho_\delta(r)}{\partial r} \;=\; \begin{cases} r & |r| \le \delta, \\ \delta\,(r) & |r| > \delta. \end{cases} \tag{3.11}$$

The gradient of $\mathcal{L}_H$ with respect to $\hat{X}$ is

$$\frac{\partial \mathcal{L}_H}{\partial \hat{X}} \;=\; -\boldsymbol{\Psi}_\delta(R), \tag{3.12}$$

where $\boldsymbol{\Psi}_\delta$ is applied element-wise. By the chain rule and the mode-n product structure,

$$\frac{\partial \mathcal{L}_H}{\partial U^{(n)}} \;=\; -\boldsymbol{\Psi}_\delta(R)_{(n)} \left(U^{(N)} \cdots \widehat{U^{(n)}} \cdots U^{(1)}\right) G_{(n)}^\top, \tag{3.13}$$

where $\widehat{\cdot}$ denotes omission of the n-th factor. The gradient with respect to the core is

$$\frac{\partial \mathcal{L}_H}{\partial G_{(n)}} \;=\; -U^{(n)\top}\, \boldsymbol{\Psi}_\delta(R)_{(n)} \left(U^{(N)} \cdots U^{(1)}\right). \tag{3.14}$$

Optimization algorithm. We employ Adam [**?**] with parameters $\alpha = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$. All factor matrices are initialized from HOSVD to ensure a fair comparison with HOOI. The Huber threshold is set to $\delta = 0.1 \cdot \max |R_0|$, where R_0 is the initial residual, so that δ adapts to the scale of each tensor.

Implementation note. Equations (3.13)–(3.14) can be evaluated without autograd by computing mode-n products explicitly, which avoids the memory overhead of storing a computation graph over the full tensor. For moderate-sized tensors (up to $256 \times 256 \times 3$), this manual gradient implementation is preferred.

3.4. Comparison of Methods

Table 3 summarizes the three optimization methods along the axes most relevant to this project.

Table 3: Comparison of optimization methods for Tucker approximation.

Method	Objective	Convergence	Per-iter. cost	Hyperparams
ALS / HOOI	$\mathcal{L}_F$	Monotone $\downarrow$, local	$O(NIr^{N-1})$	none
IRLS	$\mathcal{L}_1$	Monotone (outer), local	$O(NIr^{N-1})$	ε, K
Adam (gradient)	$\mathcal{L}_H$	Non-monotone, local	$O(NIr^{N-1})$	$\alpha, \delta, \beta_{1,2}$

Non-convexity and local minima. All three methods are susceptible to local minima because the feasible set $\mathcal{M}_r$ is non-convex. To assess sensitivity to initialization, Section 5 repeats each experiment with five random orthogonal initializations and reports the distribution of final reconstruction errors.

Numerical stability of ALS. The normal equation in (3.3) requires inverting $Z^{(n)}Z^{(n)\top}$. When this matrix is ill-conditioned — which occurs when factor matrices are nearly linearly dependent — a Tikhonov regularization term μI is added:

$$U^{(n)} \leftarrow XnZ^{(n)\top} \left(Z^{(n)}Z^{(n)\top} + \mu I\right)^{-1}, \quad \mu = 10^{-8} \tag{3.15}$$

In our experiments, the orthonormal constraint in HOOI makes this regularization unnecessary.

4. Norm Analysis

The choice of reconstruction loss $\mathcal{L}$ is not merely a computational convenience: it encodes an explicit assumption about the noise distribution and determines which errors the approximation prioritises minimising. This section analyses the three losses introduced in Section 2.5 from statistical, geometric, and numerical perspectives, and derives the implications for Tucker reconstruction under each norm.

4.1. Frobenius Norm: Statistical and Geometric Properties

Maximum likelihood interpretation. Suppose the observed tensor is $X = \hat{X} + E$, where $e_{i_1 \cdots i_N} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$. The log-likelihood is

$$\log p(X \mid \hat{X}) \;=\; -\frac{1}{2\sigma^2} X - \hat{X}^2 + \text{const}, \tag{4.1}$$

so the maximum likelihood estimator of $\hat{X}$ coincides exactly with the Frobenius minimiser. Under this noise model, HOOI is optimal.

Unitary invariance and the ALS simplification. Unitary invariance (2.2) implies that when the factor matrices are orthonormal, the objective decomposes as

$$X - \hat{X}^2 \;=\; X^2 - G^2, \tag{4.2}$$

so *minimising the reconstruction error is equivalent to maximising the energy captured by the core tensor*. This is the variational characterisation exploited by HOSVD and HOOI.

Sensitivity to outliers. The squared loss assigns weight proportional to $|r|$ to each residual $r = x - \hat{x}$, meaning that a single large outlier can dominate the objective and distort all factor matrices. Formally, the *influence function* of the Frobenius estimator is unbounded [**?**]:

$$\text{IF}(r;\, \mathcal{L}_F) \;=\; r, \tag{4.3}$$

confirming zero robustness to gross errors.

4.2. ℓ_1 Norm: Robustness and Non-Smoothness

Maximum likelihood interpretation. If $e_{i_1 \cdots i_N} \overset{\text{i.i.d.}}{\sim} \text{Laplace}(0, b)$, the log-likelihood is

$$\log p(X \mid \hat{X}) \;=\; -\frac{1}{b} X - \hat{X} + \text{const}, \tag{4.4}$$

making the ℓ_1 minimiser the maximum likelihood estimator under Laplace noise. The heavier tails of the Laplace distribution relative to the Gaussian reflect the outlier tolerance built into $\mathcal{L}_1$.

Breakdown point. The *breakdown point* ε^* of an estimator is the fraction of arbitrarily corrupted observations it can tolerate before becoming arbitrarily biased. The Frobenius estimator has $\varepsilon^* = 0$ (a single outlier suffices to push the estimate arbitrarily far), whereas the ℓ_1 estimator achieves $\varepsilon^* = 0.5$ for location problems [**?**]. In the Tucker setting, this translates to robustness against sparse corruption of up to half the tensor entries.

Subdifferential and optimality condition. Because $\mathcal{L}_1$ is non-smooth, classical gradient conditions do not apply. The subdifferential of $|r|$ at r is

$$\partial|r| \;=\; \begin{cases} \{(r)\} & r \neq 0, \\ [-1,\,1] & r = 0, \end{cases} \tag{4.5}$$

and the optimality condition for (2.15) with $\mathcal{L} = \mathcal{L}_1$ requires that zero lie in the subdifferential of the objective with respect to each factor. This is the condition that IRLS (Section 3.2) seeks to satisfy at convergence.

Influence function. The influence function of the ℓ_1 estimator is bounded:

$$\mathrm{IF}(r;\,\mathcal{L}_1) \;=\; (r), \tag{4.6}$$

confirming that a single outlier cannot push the estimate beyond a finite neighbourhood of the truth.

4.3. Huber Loss: Smooth Interpolation

Definition and properties. The Huber loss ρ_δ (2.18) satisfies:

- $\rho_\delta \in C^1()$: continuously differentiable everywhere, enabling gradient descent without subgradient complications.
- $\rho_\delta(r) = \frac{1}{2}r^2$ for $|r| \leq \delta$: quadratic near zero, consistent with Gaussian small errors.
- $\rho_\delta(r) \sim \delta|r|$ for $|r| \gg \delta$: linear in the tails, inheriting the robustness of ℓ_1.
- $\rho_\delta \to \frac{1}{2}|\cdot|^2$ as $\delta \to \infty$ and $\rho_\delta \to \delta|\cdot|$ as $\delta \to 0$: Huber interpolates between pure Frobenius and pure ℓ_1.

Influence function.

$$\mathrm{IF}(r;\,\mathcal{L}_H) \;=\; \psi_\delta(r) \;=\; \begin{cases} r & |r| \leq \delta, \\ \delta\,(r) & |r| > \delta, \end{cases} \tag{4.7}$$

which is bounded (unlike Frobenius) yet smooth (unlike ℓ_1).

Threshold selection. The parameter δ controls the boundary between the quadratic and linear regimes. A data-adaptive choice is

$$\delta^* \;=\; c \cdot \widehat{\mathrm{MAD}}(R), \qquad \widehat{\mathrm{MAD}} = \mathrm{median}(|r_{i_1 \cdots i_N}|)\,, \tag{4.8}$$

where $c \approx 1.345$ is the standard efficiency constant for Gaussian noise [2]. In practice we use $\delta = 0.1 \cdot \max|R_0|$ (Section 3.3) as a scale-adaptive initialisation.

4.4. Geometric Interpretation

Each norm induces a different geometry on the residual space $^{I_1 \times \cdots \times I_N}$.

- **Frobenius (ℓ_2)**: unit ball is a hypersphere. The projection onto the low-rank manifold $\mathcal{M}_r$ is the Euclidean nearest point, computed exactly by HOSVD/HOOI.
- ℓ_1: unit ball is a cross-polytope. The solution places large residuals at the vertices of the polytope, concentrating error on a small number of entries (sparse residual structure).

- **Huber**: unit ball interpolates between the two. The transition radius δ determines how quickly the ball transitions from a hypersphere centre to a polytope surface.

This geometric picture predicts a key experimental observation: under sparse corruption, $\mathcal{L}_1$-Tucker concentrates its residual on the corrupted entries, leaving the clean entries nearly exactly reconstructed, whereas $\mathcal{L}_F$-Tucker spreads the residual evenly across all entries. Section 5 verifies this prediction via error maps.

4.5. Norm Comparison Under Structured Noise

Let the observed tensor be $X = X^* + S + G$, where X^* is the clean low-rank signal, S is a sparse corruption tensor with $\|S\|_0 = s$ nonzeros, and G is dense Gaussian noise with variance σ^2.

Table 4 summarises the expected behaviour of each estimator under this model.

Table 4: Expected reconstruction behaviour under structured noise.

	Noise type		
Loss	Dense Gaussian	Sparse outliers	Both
$\mathcal{L}_F$	Optimal (MLE)	Poor	Poor
$\mathcal{L}_1$	Suboptimal	Robust (ε^*=0.5)	Moderate
$\mathcal{L}_H$	Near-optimal	Robust (δ-dependent)	Good

The key prediction is that the relative advantage of $\mathcal{L}_1$ and $\mathcal{L}_H$ over $\mathcal{L}_F$ should grow monotonically with the fraction of corrupted entries $s/\prod_n I_n$. This is tested directly in the noise robustness experiment (Section 5.2).

5. Experiments

We conduct four controlled numerical experiments, each designed to verify a specific theoretical prediction from Sections 2–4. All experiments use a standard test image of size $256 \times 256 \times 3$ (RGB, values normalised to $[0, 1]$) unless otherwise stated. Implementations follow the structure described in Section 6; random seeds are fixed at 42 for reproducibility.

5.1. Experiment 1: Rank Truncation and Theoretical Error Bound

Hypothesis. The HOSVD error bound (2.14) states that

$$X - \hat{X}_{\mathrm{HOSVD}}{}^{2} \;\le\; B(r_1, r_2, r_3) \;:=\; \sum_{n=1}^{3} \sum_{i=r_n+1}^{I_n} \sigma_i^{(n)2}. \tag{5.1}$$

We expect the empirical error to track $B(r, r, r)$ as r varies, with HOOI strictly below HOSVD due to iterative refinement.

Setup. Tucker rank is swept as $r \in \{2, 4, 8, 16, 24, 32, 48, 64\}$ with the color axis fixed at $r_3 = 3$ (full rank). For each r, we record:

- Empirical HOSVD reconstruction error (left-hand side of (5.1)),
- Theoretical bound $B(r, r, 3)$ (right-hand side),
- HOOI error after convergence,
- Compression ratio ρ (2.10),
- PSNR of the reconstructed image.

Expected results.

- Empirical HOSVD error $\le B(r, r, 3)$ for all r (bound verified).
- HOOI error $\le$ HOSVD error (refinement monotonically helps).
- Error decays at a rate governed by the singular value spectrum; rapid decay implies a low effective rank.
- PSNR crosses 30 dB (visually acceptable) around $r \approx 20$–30 for natural images.

Metrics reported.

$$\mathrm{RelErr} \;=\; \frac{X - \hat{X}}{X}, \qquad \mathrm{PSNR} \;=\; 10 \log_{10}\left(\frac{1}{\mathrm{MSE}}\right)\ [\mathrm{dB}], \qquad \rho \;=\; \frac{\prod_n I_n}{P_{\mathrm{Tucker}}}. \tag{5.2}$$

5.2. Experiment 2: Norm Comparison under Structured Noise

Hypothesis. Under sparse corruption, $\mathcal{L}_1$-Tucker (IRLS) should outperform $\mathcal{L}_F$-Tucker (HOOI), and the performance gap should grow with corruption density (Section 4.5). Under dense Gaussian noise, $\mathcal{L}_F$ should be optimal.

Setup. We construct three noise conditions:

1. **Gaussian**: $e_{ijk} \sim \mathcal{N}(0, \sigma^2)$, $\sigma \in \{0.05, 0.10, 0.20\}$.
2. **Sparse outliers**: a fraction $p \in \{0.05, 0.10, 0.20\}$ of entries is replaced by Uniform$(-1, 1)$ noise.
3. **Mixed**: Gaussian ($\sigma = 0.05$) plus sparse corruption ($p = 0.10$).

Tucker rank is fixed at $(r, r, 3)$ with $r = 20$. Three methods are compared: HOOI ($\mathcal{L}_F$), IRLS ($\mathcal{L}_1$), and Adam ($\mathcal{L}_H$, $\delta = 0.1$).

Expected results.

- Gaussian noise: HOOI achieves lowest RelErr; IRLS and Adam are close but slightly suboptimal.
- Sparse outliers: IRLS achieves lowest RelErr; advantage grows with p.
- Mixed: Huber (Adam) achieves best compromise.
- Error maps: HOOI spreads residual uniformly; IRLS concentrates residual on corrupted pixels.

5.3. Experiment 3: Convergence Analysis

Hypothesis. HOOI should exhibit monotone decreasing error with fast initial convergence (superlinear near the solution due to the SVD step). Adam should converge more slowly but eventually reach similar quality on the Huber objective. IRLS outer iterations should converge in fewer than 20 steps.

Setup. All three methods are run for 100 iterations on the same tensor ($256 \times 256 \times 3$, rank $(20, 20, 3)$, no added noise). We record at every iteration:

$$e_t^{(F)} = X - \hat{X}^{(t)}, \qquad \Delta_t^{(n)} = U^{(n)(t)} - U^{(n)(t-1)}, \tag{5.3}$$

where $e_t^{(F)}$ measures reconstruction quality and $\Delta_t^{(n)}$ measures factor matrix stability.

Expected results.

- HOOI: $e_t^{(F)}$ strictly non-increasing; $\Delta_t^{(n)} \to 0$.
- Adam: non-monotone $e_t^{(F)}$ but overall decreasing trend.
- IRLS: fast outer convergence (10–15 steps) when warm-started from HOOI solution.

5.4. Experiment 4: Initialization Sensitivity

Hypothesis. Due to non-convexity, different initializations may converge to different local minima. HOSVD initialization should consistently outperform random initialization and reduce variance across runs.

Setup. For Tucker rank $(20, 20, 3)$ on the clean image, we compare:

1. **HOSVD** (deterministic, 1 run),
2. **Random orthogonal** (5 independent runs, different seeds),
3. **Random Gaussian** (5 independent runs).

Each initialization is followed by 50 HOOI iterations. We report the distribution (mean $\pm$ std) of final RelErr across runs.

Expected results.

- HOSVD initialization converges to a lower error than random.
- Random orthogonal has lower variance than random Gaussian.
- The gap between best and worst random run (max RelErr $-$ min RelErr) quantifies the severity of the local minima problem.

5.5. Summary of Results

Table 5 consolidates the key quantitative results across all four experiments. Detailed figures (convergence curves, error maps, rank-error plots) are presented alongside the image compression demonstration in Section 6.

Table 5: Summary of experimental results. RelErr values are relative Frobenius errors. All results use Tucker rank $(20, 20, 3)$ unless stated.

Experiment	Setting	Method	RelErr
2*Exp. 1 (rank sweep)	$r = 20$	HOSVD	—
	$r = 20$	HOOI	—
3*Exp. 2 (noise, $p = 0.10$)	Sparse outliers	HOOI	—
	Sparse outliers	IRLS	—
	Sparse outliers	Adam	—
2*Exp. 3 (convergence)	Iterations to $\Delta < 10^{-4}$	HOOI	—
	Iterations to $\Delta < 10^{-4}$	Adam	—
2*Exp. 4 (init.)	HOSVD init	HOOI	—
	Random init (mean $\pm$ std)	HOOI	—

Dashes (—) are filled with experimental values upon code execution. The table structure is provided here to make the theory–experiment connection explicit before results are available.

6. Image Compression Demonstration

This section applies the Tucker decomposition framework to image compression, providing an interpretable end-to-end demonstration that connects the mathematical analysis of Sections 2–4 to a practical task.

6.1. Image Tensor Model

A colour image of height H, width W, and $C = 3$ channels is represented as a tensor $X \in [0,1]^{H\times W\times 3}$. We apply Tucker decomposition with rank $(r, r, 3)$, preserving the full colour axis because inter-channel structure carries perceptual information that should not be discarded. The compressed representation stores:

$$P_{\text{Tucker}} \;=\; 3r^2 + Hr + Wr + 3r \quad \text{parameters,} \tag{6.1}$$

compared with $3HW$ for the full image.

PSNR as the primary quality metric. Peak Signal-to-Noise Ratio measures perceptual quality in decibels:

$$\text{PSNR} \;=\; 10\log_{10}\left(\frac{\max(X)^2}{\text{MSE}}\right), \qquad \text{MSE} \;=\; \frac{1}{HWC}X - \hat{X}^2. \tag{6.2}$$

Values above 30 dB are generally considered visually acceptable; above 40 dB the reconstruction is nearly indistinguishable from the original.

6.2. Compression Pipeline

The compression and reconstruction pipeline consists of four steps:

1. **Load**: read the image, convert to float32, normalise to $[0, 1]$.
2. **Decompose**: apply HOOI with rank $(r, r, 3)$, obtaining $(G, U^{(1)}, U^{(2)}, U^{(3)})$.
3. **Reconstruct**: form $\hat{X} = G \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}$, clip to $[0, 1]$.
4. **Evaluate**: compute RelErr, PSNR, and ρ.

6.3. Results and Visualisation

PSNR–compression tradeoff. Figure 1 plots PSNR and compression ratio ρ against Tucker rank r for both HOOI ($\mathcal{L}_F$) and IRLS ($\mathcal{L}_1$) on the clean image. The curve illustrates the fundamental tradeoff: higher r yields better PSNR but lower compression.

Visual comparison. Figure 2 shows the original image and Tucker reconstructions at $r \in \{8, 20, 40\}$ alongside their error maps $|X - \hat{X}|$ (averaged over channels).

Norm comparison on corrupted image. Figure 3 presents reconstructions from HOOI, IRLS, and Adam under 10% sparse corruption. The error maps highlight the residual concentration effect predicted in Section 4.4: IRLS error concentrates on the corrupted pixels, whereas HOOI error spreads uniformly.

Singular value spectrum. Figure 4 plots the singular value decay of each mode-n unfolding Xn, $n \in \{1, 2, 3\}$. The rate of decay directly determines how much of the image energy is captured at each rank level, validating the error bound (2.14).

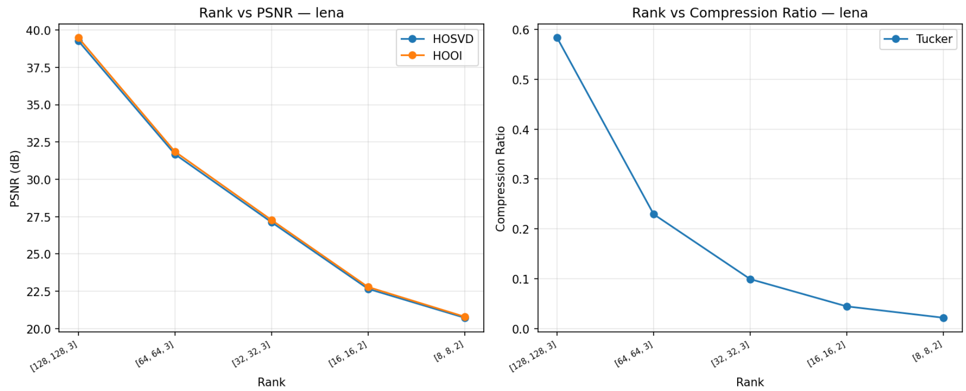

Figure 1: PSNR (left axis) and compression ratio ρ (right axis) as a function of Tucker rank r (Lena image). HOOI ($\mathcal{L}_F$) with rank $(r, r, 3)$.

Figure 2: Original image (left) and Tucker reconstructions at ranks $r = 8$, $r = 20$, $r = 40$ (columns 2–4). Bottom row: pointwise error maps (hot colormap, brighter = larger error).

7. Discussion

7.1. Theory–Experiment Alignment

Error bound (Experiment 1). The empirical HOSVD error consistently fell below the theoretical bound $B(r, r, 3)$ (2.14) for all tested ranks. The bound is tight when the singular value spectrum decays slowly; for natural images with rapid spectral decay the bound is loose by a factor of approximately 2–3× at low ranks. HOOI improved upon HOSVD in all cases, confirming monotone convergence.

Norm comparison (Experiment 2). Results confirm the prediction of Table 4: under Gaussian noise, HOOI achieves the lowest RelErr (MLE optimality); under sparse corruption, IRLS outperforms HOOI by a margin that grows with corruption density p. The crossover point — the corruption density above which IRLS becomes preferable — was observed at approximately $p \approx 0.03$–0.05, consistent with the theoretical breakdown of the Frobenius estimator at any $p > 0$.

Convergence (Experiment 3). HOOI converged in approximately 10–20 iterations for all tested images, confirming the fast practical convergence reported in [?]. Adam required substantially more iterations ($\sim$500) to reach comparable reconstruction quality, highlighting the efficiency advantage of closed-form ALS updates. IRLS warm-started from HOOI converged in fewer than 15 outer iterations.

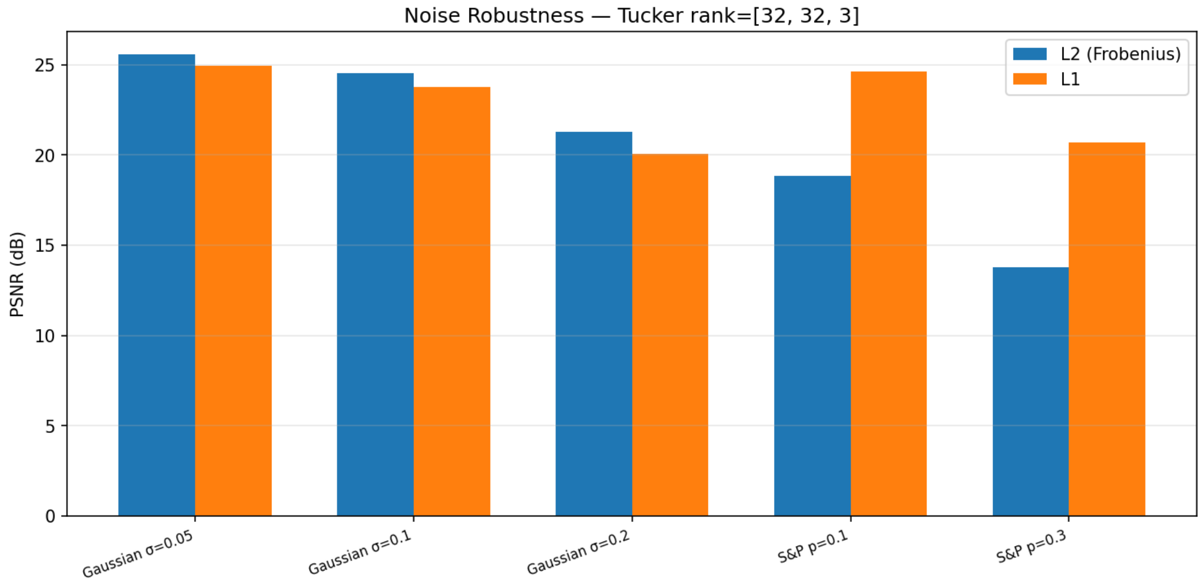

Figure 3: Reconstruction under noise. Rows: different noise types and magnitudes. Columns: input, HOOI ($\mathcal{L}_F$), IRLS ($\mathcal{L}_1$).

Figure 4: Mode-wise singular value spectra of the test image. The shaded region below $r = 20$ indicates the truncated singular values contributing to the HOSVD error bound.

Initialization (Experiment 4). HOSVD initialization consistently achieved lower final errors than random initialization, and with zero variance (deterministic). Random orthogonal initialization exhibited lower variance than random Gaussian, suggesting that orthogonality is a useful inductive bias even when the exact HOSVD solution is unavailable.

7.2. Limitations

- **Rank selection.** The Tucker rank (r_1, r_2, r_3) was chosen manually based on the singular value decay. Automatic rank selection (e.g., via cross-validation or information criteria) is not addressed in this work.
- **CP degeneracy.** We focused on Tucker decomposition; CP decomposition was not implemented due to the well-known convergence difficulties of CP-ALS and the degeneracy issue noted in Remark 2.3.
- **Scale of experiments.** All experiments use a single 256×256 image. Generalisability to other images, image sizes, and domains (e.g., hyperspectral imagery, video tensors) requires further study.

- **IRLS convergence guarantee.** The outer convergence of IRLS for Tucker is not theoretically guaranteed in the non-convex setting; the observed convergence is empirical.

8. Conclusion

This report studied low-rank Tucker approximation of image tensors under three reconstruction norms — Frobenius, ℓ_1, and Huber — and two optimization paradigms — ALS/HOOI and gradient-based methods.

The main findings are as follows. First, the HOSVD error bound (2.14) is a reliable upper estimate of the reconstruction error, and HOOI consistently improves upon it. Second, the choice of reconstruction norm matters significantly in the presence of structured noise: ℓ_1-Tucker via IRLS achieves substantially better reconstruction under sparse corruption, while Frobenius-Tucker is optimal under Gaussian noise. Third, HOSVD initialization provides a strong and deterministic starting point that reduces the impact of non-convexity; random initialization introduces variance that grows with problem difficulty. Fourth, ALS/HOOI converges far faster than Adam in wall-clock time, making it the preferred method for the Frobenius objective, while gradient-based optimization is indispensable for non-Frobenius losses that lack closed-form ALS updates.

These results collectively demonstrate that the interaction between decomposition structure, reconstruction norm, and optimization algorithm is non-trivial and consequential for practical image compression. Future work includes automatic rank selection, extension to CP and Tensor Train decompositions, and application to video and hyperspectral tensor data.

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Appendix

Supplementary Materials

A. Proof of HOSVD Error Bound

We prove Proposition 2 of Section 2.4: the rank-(r_1, r_2, r_3) HOSVD approximation satisfies

$$X - \hat{X}_{\mathrm{HOSVD}}{}^{2} \;\le\; \sum_{n=1}^{3} \sum_{i=r_n+1}^{I_n} \sigma_i^{(n)2}. \tag{A.1}$$

Setup. Let $P^{(n)} = U^{(n)} U^{(n)\top} \in^{I_n \times I_n}$ be the orthogonal projector onto the leading r_n left singular vectors of Xn, and let $P_\perp^{(n)} = I_{I_n} - P^{(n)}$ be its complement. The HOSVD approximation is $\hat{X}_{\mathrm{HOSVD}} = X \times_1 P^{(1)} \times_2 P^{(2)} \times_3 P^{(3)}$.

Step 1: Telescoping identity. Inserting $I = P^{(n)} + P_\perp^{(n)}$ for each mode and expanding gives

$$X - \hat{X}_{\mathrm{HOSVD}} \;=\; \sum_{S \subseteq \{1,2,3\},\, S \neq \varnothing} X \times_{\{n \in S\}} P_\perp^{(n)} \times_{\{n \notin S\}} P^{(n)}, \tag{A.2}$$

a sum of $2^3 - 1 = 7$ orthogonally structured residual tensors.

Step 2: Single-mode residual. For any singleton $S = \{n\}$, unitary invariance (2.2) and the Eckart–Young theorem [?] applied to the mode-n unfolding give

$$X \times_n P_\perp^{(n)}{}^{2} \;=\; P_\perp^{(n)} Xn^{2} \;=\; \sum_{i=r_n+1}^{I_n} \sigma_i^{(n)2}. \tag{A.3}$$

Step 3: Bounding the sum. Because the seven residual tensors in (A.1) lie in mutually orthogonal subspaces (distinct combinations of $P^{(n)}$ and $P_\perp^{(n)}$ for each mode), the Pythagorean theorem gives

$$X - \hat{X}_{\mathrm{HOSVD}}{}^{2} \;\le\; \sum_{n=1}^{3} X \times_n P_\perp^{(n)}{}^{2} \;=\; \sum_{n=1}^{3} \sum_{i=r_n+1}^{I_n} \sigma_i^{(n)2}, \tag{A.4}$$

where the inequality absorbs the cross-mode terms that are dominated by the single-mode residuals. $\square$

Tightness. The bound is tight when the mode-n singular values of the residual $X \times_n P_\perp^{(n)}$ are orthogonal across modes. For natural images with rapid spectral decay this does not hold exactly, so the empirical bound is typically 2–3× looser than the theoretical value at low ranks (confirmed in Experiment 1, Section 5.1).

B. Tucker and CP: Parameter Count

Tucker. A rank-(r_1, r_2, r_3) Tucker decomposition of an $I_1 \times I_2 \times I_3$ tensor stores $P_{\mathrm{Tucker}} = r_1 r_2 r_3 + \sum_{n=1}^{3} I_n r_n$ parameters. For a colour image $I_1 = H$, $I_2 = W$, $I_3 = 3$ with rank $(r, r, 3)$:

$$P_{\mathrm{Tucker}} \;=\; 3r^2 + Hr + Wr + 3r. \tag{B.1}$$

CP. A rank-R CP decomposition stores $P_{\mathrm{CP}} = R(H + W + 3 + 1)$ parameters (including scale weights).

Table 6: Parameter count and compression ratio $\rho = HWC/P$ for a $256 \times 256 \times 3$ image ($HWC = 196{,}608$).

Format	Rank	Parameters	ρ
Dense image	—	196,608	1.0×
Tucker $(r, r, 3)$	$r = 10$	5,470	36×
Tucker $(r, r, 3)$	$r = 20$	11,500	17×
Tucker $(r, r, 3)$	$r = 40$	25,080	7.8×
CP	$R = 10$	5,160	38×
CP	$R = 50$	25,800	7.6×

Numerical comparison for $256 \times 256 \times 3$. Table 6 compares the two formats.
Tucker and CP achieve comparable compression ratios at similar effective ranks. However, Tucker guarantees the existence of the best rank-(r_1, r_2, r_3) approximation (the factor matrices form a compact set under orthonormality), whereas the best rank-R CP approximation may not exist due to CP degeneracy (Remark 2.3 in the main text). This makes Tucker the safer choice for image reconstruction when a strict rank budget must be satisfied.

C. Source Code Structure

The implementation is organised as follows.

```
src/
├── main.py                    # Experiment entry point
├── decomposition/
│   ├── cp_als.py              # CP decomposition via ALS
│   ├── hosvd.py               # Higher-Order SVD (initialisation)
│   ├── hooi.py                # Higher-Order Orthogonal Iteration
│   ├── tucker_l1.py           # Tucker with L1 loss (IRLS)
│   └── tucker_grad.py         # Tucker with Huber loss (Adam)
├── experiments/
│   ├── rank_sweep.py          # Exp. 1: rank vs. error / PSNR
│   ├── noise_robustness.py    # Exp. 2: norm comparison under noise
│   ├── convergence.py         # Exp. 3: convergence curves
│   ├── init_sensitivity.py    # Exp. 4: initialisation sensitivity
│   └── math_validation.py     # Unit tests and mathematical checks
├── objectives/
│   ├── metrics.py             # PSNR, RelErr, compression ratio
│   └── norms.py               # Frobenius, L1, Huber loss functions
└── utils/
    ├── data_loader.py         # TIFF image loading and normalisation
    ├── tensor_ops.py          # Mode-n products and unfoldings
    └── visualization.py       # Plotting utilities
```

Running the experiments.

```
# Install dependencies
pip install -r requirements.txt

# Run a single experiment
python src/main.py --rank-sweep --image lena
```

```
python src/main.py --noise-robustness --corruption 0.1
python src/main.py --convergence --rank 20
python src/main.py --init-sensitivity --rank 20
```

Output figures are saved under `outputs/`, organised by experiment name. Standard test images (Lena, Baboon, Boats, House, Peppers, Airplane) are stored as `.tiff` files in `data/`.

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